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D20A - 40

Experiment: 08

Aim: To implement the Blockchain platform Ganache and deploy a simple Solidity smart contract using Metamask and Remix IDE.

Theory:
Ganache is a personal Ethereum blockchain used for testing and development. It allows developers to deploy contracts, develop decentralized applications (DApps), and run tests in a safe and deterministic environment.

<u>Ganache provides:</u>

- Instant mining of blocks
- Pre-funded test accounts
- Transaction history view
- Graphical interface to monitor blockchain activities

It is part of the Truffle Suite and is commonly used with Remix and Metamask for smart contract deployment.

<u>Connection between Ganache, MetaMask, and Remix IDE</u>

These three tools work together to create a complete blockchain workflow:

1. Ganache → MetaMask

- Ganache provides **local blockchain + accounts**
- MetaMask connects using:
    - RPC URL (e.g., 127.0.0.1:7545)
- Ganache accounts are imported into MetaMask using private keys
- MetaMask can now use Ganache accounts

2. MetaMask → Remix IDE

- In Remix, select:
  👉 **Injected Provider - MetaMask**
- Remix connects to MetaMask wallet
- Remix uses MetaMask account to perform transactions

3. Remix IDE → Ganache

- When you deploy a contract:
    - Remix → sends request to MetaMask
    - MetaMask → signs transaction
    - Ganache → executes transaction

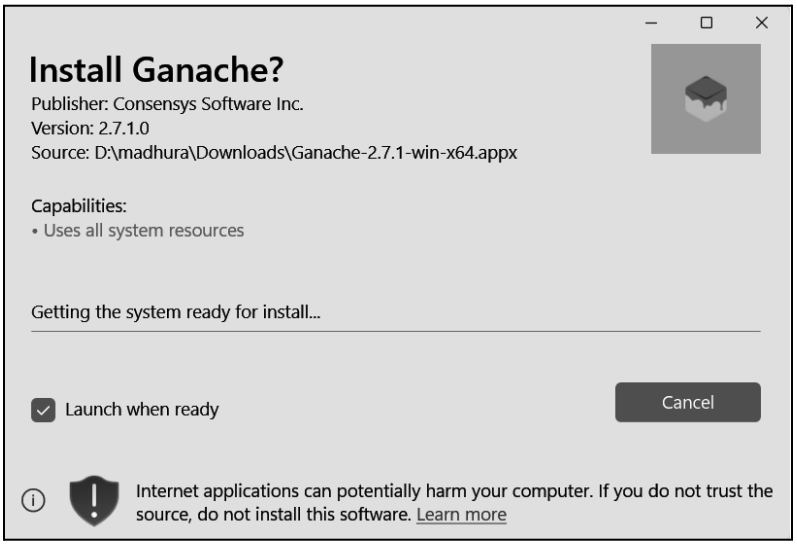
The contract gets deployed on the local blockchain.

Advantages of this setup

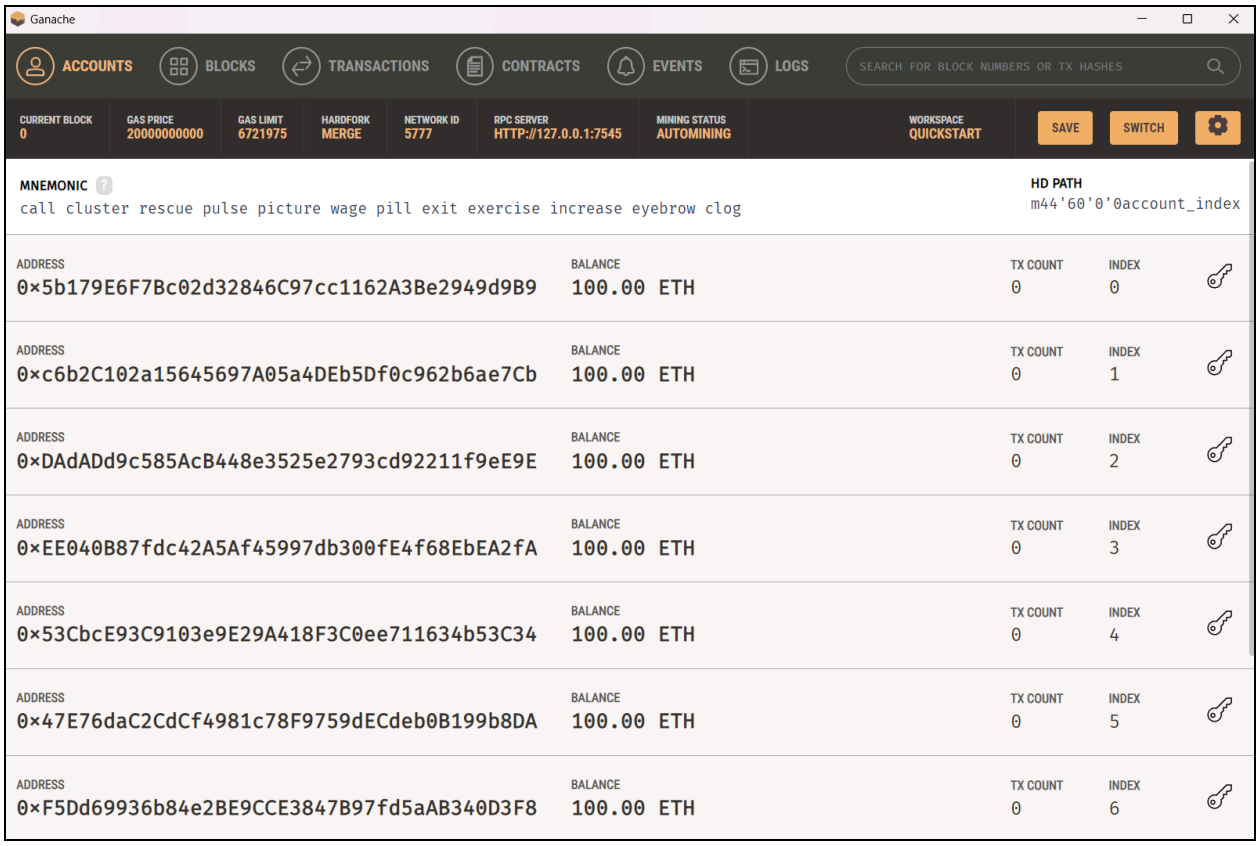
- No real money required
- Fast execution
- Easy debugging
- Safe testing environment

Implementation:

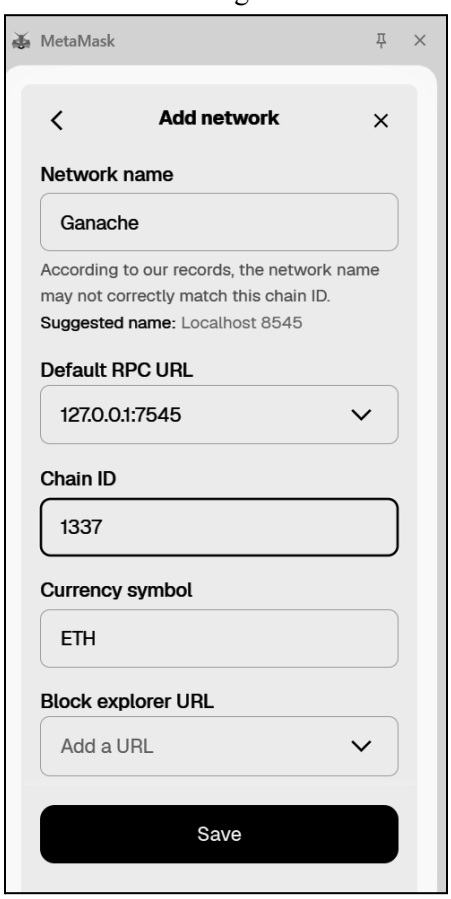
Installing Ganache in system-



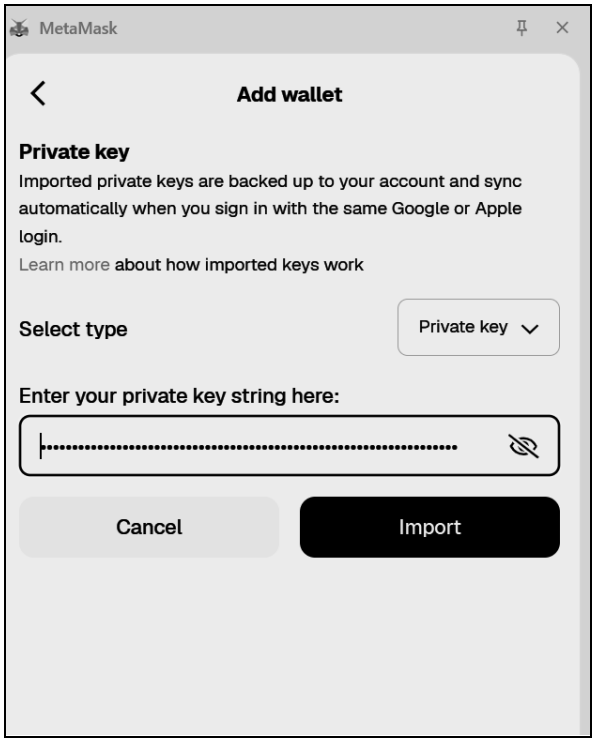
After installing Ganache, clicking on quickstart we get 10 accounts with 100ETH each-



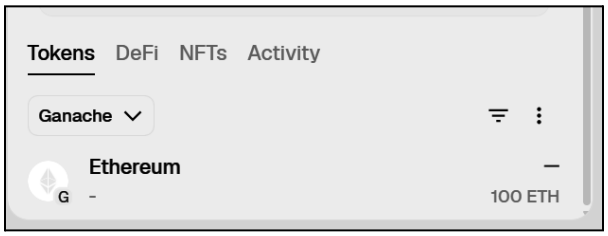
In Matamask Adding Ganache network-



Copying the private key of account 0 and importing that account in metamask-

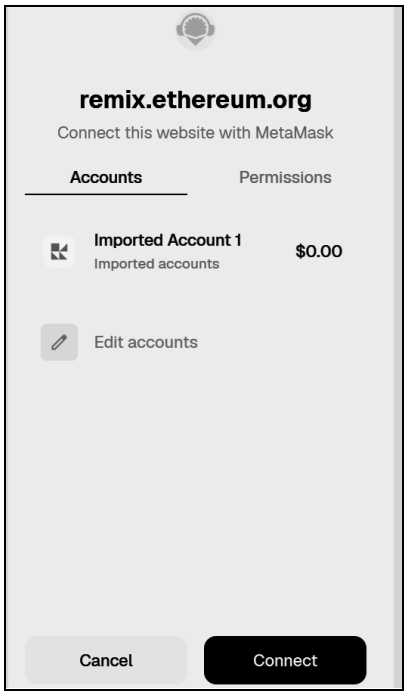


Account imported successfully balance is 100 ETH-



Selecting injected provider - metamask environment in remix ide-

Connection with remix ide-

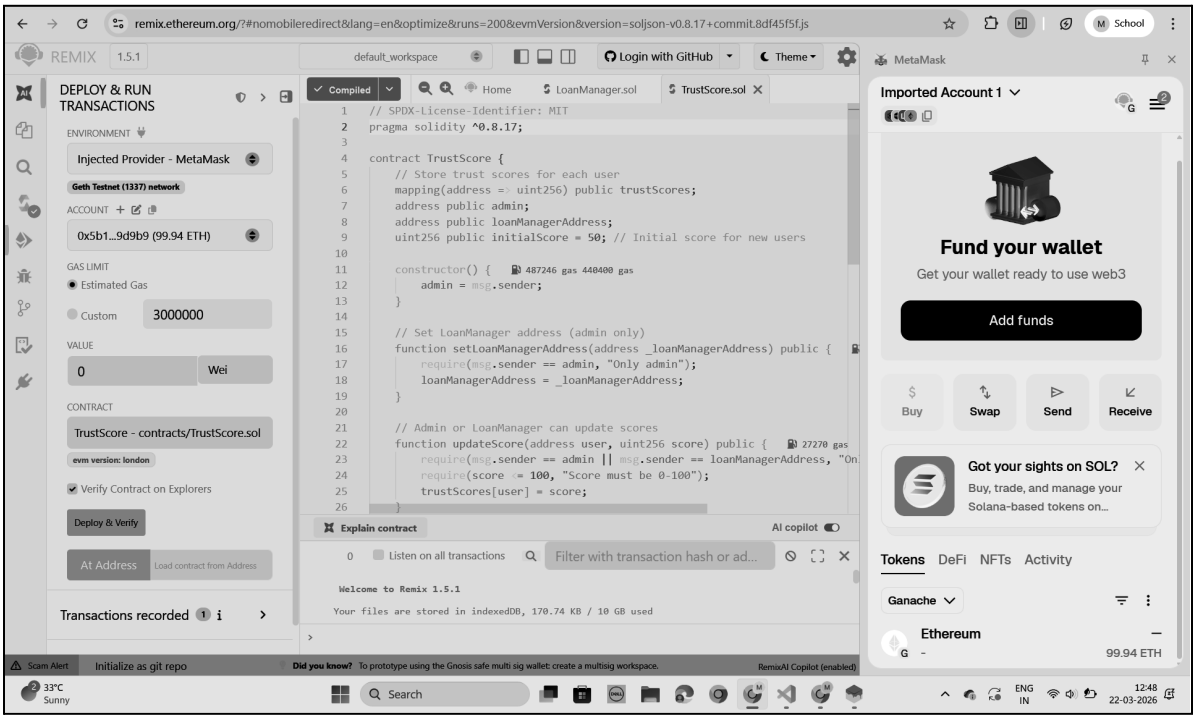


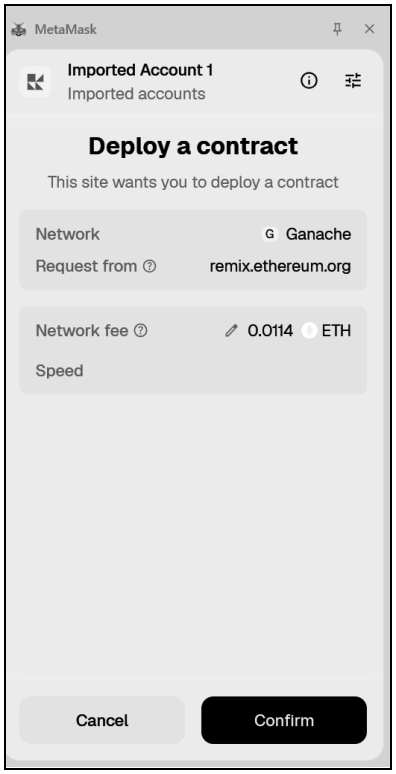
Deploying contracts-

Contract 1- TrustScore.sol

```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.17;

contract TrustScore {
    // Store trust scores for each user
    mapping(address => uint256) public trustScores;
    address public admin;
    address public loanManagerAddress;
    uint256 public initialScore = 50; // Initial score for new users

    constructor() {
        admin = msg.sender;
    }
```

```
    // Set LoanManager address (admin only)
    function setLoanManagerAddress(address
_loanManagerAddress) public {
        require(msg.sender == admin, "Only admin");
        loanManagerAddress = _loanManagerAddress;
    }

    // Admin or LoanManager can update scores
    function updateScore(address user, uint256 score)
public {
        require(msg.sender == admin || msg.sender ==

loanManagerAddress, "Only admin or
LoanManager");
        require(score <= 100, "Score must be 0-100");
        trustScores[user] = score;
    }

    // Set initial score for new users (admin only)
    function setInitialScore(uint256 _initialScore)
public {
        require(msg.sender == admin, "Only admin");
        require(_initialScore <= 100, "Score must be
0-100");
        initialScore = _initialScore;
    }

    // Anyone can check a score - returns initialScore if
not set
    function getScore(address user) public view
returns (uint256) {
        uint256 score = trustScores[user];
        // If score is 0, return initial score (user hasn't
been explicitly set)
        return score > 0 ? score : initialScore;
    }
}
```





Contract 2- LoanManager.sol

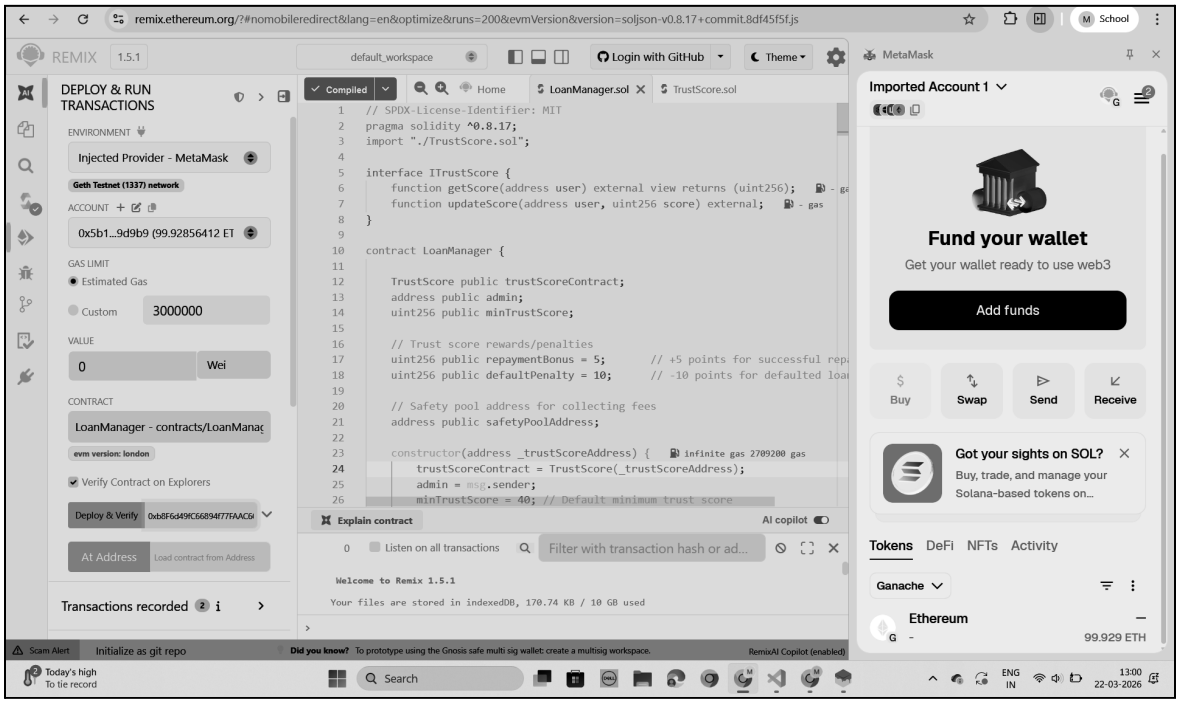
```
// SPDX-License-Identifier: MIT
pragma solidity ^0.8.17;
import "./TrustScore.sol";

interface ITrustScore {
    function getScore(address user) external view
returns (uint256);
    function updateScore(address user, uint256 score)
external;
}

contract LoanManager {

    TrustScore public trustScoreContract;
    address public admin;
    uint256 public minTrustScore;

    // Trust score rewards/penalties
    uint256 public repaymentBonus = 5;       // +5
points for successful repayment
    uint256 public defaultPenalty = 10;      // -10 points
for defaulted loan

    // Safety pool address for collecting fees
    address public safetyPoolAddress;

    constructor(address _trustScoreAddress) {
        trustScoreContract =
TrustScore(_trustScoreAddress);
        admin = msg.sender;
        minTrustScore = 40; // Default minimum trust
score
    }

    // Set safety pool address (admin only)
    function setSafetyPoolAddress(address
_safetyPoolAddress) public {
        require(msg.sender == admin, "Only admin");
        safetyPoolAddress = _safetyPoolAddress;
    }

    struct Loan {
        address borrower;
        address lender;
        uint256 amount;
        uint256 interestRate; // Interest rate in basis
points (e.g., 500 = 5%)
        uint256 safetyFee;    // Safety fee in basis points
        uint256 duration;
        uint256 createdAt;
        bool funded;
        bool repaid;
        bool withdrawn;
        bool defaulted;
```

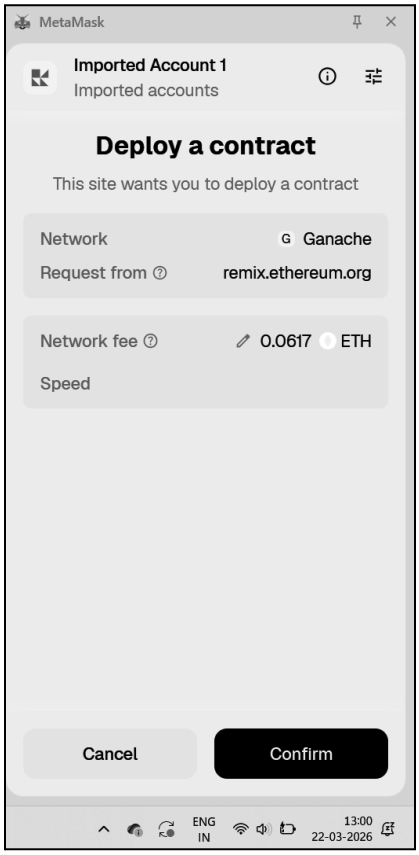
```
    }

    Loan[] public loans;

    event LoanRequested(uint256 loanId, address
borrower, uint256 amount, uint256 interestRate);
    event LoanFunded(uint256 loanId, address lender);
    event LoanWithdrawn(uint256 loanId);
    event LoanRepaid(uint256 loanId, uint256
totalRepaid, uint256 interest);
    event LoanDefaulted(uint256 loanId, address
borrower);
    event TrustScoreIncreased(address borrower,
uint256 amount);
    event TrustScoreDecreased(address borrower,
uint256 amount);
    event MinTrustScoreUpdated(uint256 newScore);

    // Calculate interest rate based on trust score
    // Higher trust score = lower interest rate
    function getInterestRate(uint256 trustScore) public
pure returns (uint256) {
        if (trustScore >= 90) return 300;      // 3% for
excellent credit
        if (trustScore >= 75) return 500;      // 5% for
good credit
        if (trustScore >= 60) return 800;      // 8% for fair
credit
        if (trustScore >= 50) return 1200;     // 12% for
poor credit
        return 1500;                           // 15% for very
poor credit
    }

    // Get maximum borrow limit based on trust score
(in wei)
    // Higher trust = larger loans allowed
    function getMaxBorrowLimit(uint256 trustScore)
public pure returns (uint256) {
        if (trustScore >= 90) return 10 ether;    //
Excellent: up to 10 ETH
        if (trustScore >= 75) return 5 ether;     // Good:
up to 5 ETH
        if (trustScore >= 60) return 2 ether;     // Fair: up
to 2 ETH
        if (trustScore >= 50) return 0.5 ether;   // Poor:
up to 0.5 ETH
        return 0.1 ether;                         // Very poor: up
to 0.1 ETH
    }

    // Get safety fee based on trust score (in basis
points)
    // Lower trust = higher safety fee to protect lenders
    function getSafetyFee(uint256 trustScore) public
pure returns (uint256) {
        if (trustScore >= 90) return 50;       // 0.5% for
excellent credit
        if (trustScore >= 75) return 100;      // 1% for
good credit
        if (trustScore >= 60) return 200;      // 2% for fair
credit
        if (trustScore >= 50) return 400;      // 4% for
poor credit
        return 600;                            // 6% for very poor
credit
    }

    // Get borrower's current limit info
    function getBorrowerLimits(address borrower)
public view returns (
        uint256 trustScore,
        uint256 maxBorrowLimit,
        uint256 interestRate,
        uint256 safetyFee
    ) {
        trustScore =
trustScoreContract.getScore(borrower);
        maxBorrowLimit =
getMaxBorrowLimit(trustScore);
        interestRate = getInterestRate(trustScore);
        safetyFee = getSafetyFee(trustScore);
    }

    // Calculate total repayment amount (principal +
interest + safety fee)
    function calculateRepayment(uint256 loanId)
public view returns (uint256) {
        Loan memory loan = loans[loanId];
        uint256 interest = (loan.amount *
loan.interestRate) / 10000;
        uint256 safetyFeeAmount = (loan.amount *
loan.safetyFee) / 10000;
        return loan.amount + interest +
safetyFeeAmount;
    }

    // Get breakdown of repayment components
    function getRepaymentBreakdown(uint256 loanId)
public view returns (
        uint256 principal,
        uint256 interest,
        uint256 safetyFeeAmount,
        uint256 total
    ) {
        Loan memory loan = loans[loanId];
```

```
        principal = loan.amount;
        interest = (loan.amount * loan.interestRate) /
10000;
        safetyFeeAmount = (loan.amount *
loan.safetyFee) / 10000;
        total = principal + interest + safetyFeeAmount;
    }

    // Admin function to update minimum trust score
    function setMinTrustScore(uint256 _minScore)
public {
        require(msg.sender == admin, "Only admin can
set minimum trust score");
        require(_minScore <= 100, "Score must be
0-100");
        minTrustScore = _minScore;
        emit MinTrustScoreUpdated(_minScore);
    }

    // Set trust score reward for repayment
    function setRepaymentBonus(uint256 _bonus)
public {
        require(msg.sender == admin, "Only admin");
        repaymentBonus = _bonus;
    }

    // Set trust score penalty for default
    function setDefaultPenalty(uint256 _penalty)
public {
        require(msg.sender == admin, "Only admin");
        defaultPenalty = _penalty;
    }

    // Check if a loan is overdue
    function isLoanOverdue(uint256 loanId) public
view returns (bool) {
        Loan memory loan = loans[loanId];
        if (!loan.withdrawn || loan.repaid ||
loan.defaulted) {
            return false;
        }
        return block.timestamp > loan.createdAt +
loan.duration;
    }

    // Mark loan as defaulted (called by admin or
lender)
    function markLoanDefaulted(uint256 loanId)
public {
        Loan storage loan = loans[loanId];
        require(!loan.defaulted && !loan.repaid, "Loan
already finalized");
        require(isLoanOverdue(loanId), "Loan is not
overdue");
        require(msg.sender == loan.lender || msg.sender
== admin, "Only lender or admin can mark default");

        loan.defaulted = true;

        // Decrease borrower's trust score
        uint256 currentScore =
trustScoreContract.getScore(loan.borrower);
        uint256 newScore = currentScore >
defaultPenalty ? currentScore - defaultPenalty : 0;
        trustScoreContract.updateScore(loan.borrower,
newScore);

        emit LoanDefaulted(loanId, loan.borrower);
        emit TrustScoreDecreased(loan.borrower,
defaultPenalty);
    }

    // ------------------------------
    // BORROWER CREATES LOAN REQUEST
    // ------------------------------
    function createLoan(uint256 amount, uint256
duration) public {

        uint256 score =
trustScoreContract.getScore(msg.sender);
        require(score >= minTrustScore, "Trust score
too low");

        // Enforce maximum borrow limit based on trust
score
        uint256 maxLimit =
getMaxBorrowLimit(score);
        require(amount <= maxLimit, "Amount exceeds
your borrow limit");

        uint256 interestRate = getInterestRate(score);
        uint256 safetyFee = getSafetyFee(score);

        loans.push(
            Loan({
                borrower: msg.sender,
                lender: address(0),
                amount: amount,
                interestRate: interestRate,
                safetyFee: safetyFee,
                duration: duration,
                createdAt: block.timestamp,
                funded: false,
                repaid: false,
                withdrawn: false,
```

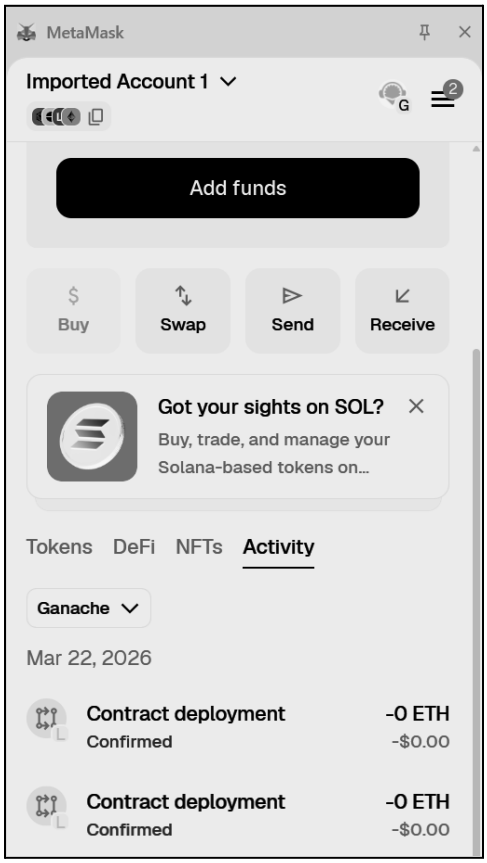
```
            defaulted: false
        })
    );

    emit LoanRequested(loans.length - 1,
msg.sender, amount, interestRate);
  }

  // ------------------------------
  // LENDER FUNDS LOAN
  // ------------------------------
  function fundLoan(uint256 loanId) public payable
{

    Loan storage loan = loans[loanId];

    require(!loan.funded, "Already funded");
    require(msg.value == loan.amount, "Incorrect
ETH amount");

    loan.lender = msg.sender;
    loan.funded = true;

    emit LoanFunded(loanId, msg.sender);
  }

  // ------------------------------
  // BORROWER WITHDRAWS FUNDS
  // ------------------------------
  function withdrawLoan(uint256 loanId) public {

    Loan storage loan = loans[loanId];

    require(msg.sender == loan.borrower, "Not
borrower");
    require(loan.funded, "Loan not funded");
    require(!loan.withdrawn, "Already withdrawn");

    loan.withdrawn = true;

    payable(msg.sender).transfer(loan.amount);

    emit LoanWithdrawn(loanId);
  }

  // ------------------------------
  // BORROWER REPAYS LOAN
  // ------------------------------
  function repayLoan(uint256 loanId) public payable
{

    Loan storage loan = loans[loanId];

    require(msg.sender == loan.borrower, "Not
borrower");
    require(loan.withdrawn, "Loan not withdrawn");
    require(!loan.repaid, "Already repaid");
    require(!loan.defaulted, "Loan is defaulted");
    require(loan.lender != address(0), "Invalid
lender address");

    // Get repayment breakdown
    (uint256 principal, uint256 interest, uint256
safetyFeeAmount, uint256 totalRepayment) =
getRepaymentBreakdown(loanId);
    require(msg.value >= totalRepayment,
"Insufficient repayment amount");

    // Transfer principal + interest to lender
    uint256 lenderAmount = principal + interest;
    payable(loan.lender).transfer(lenderAmount);

    // Transfer safety fee to safety pool (if set)
    if (safetyPoolAddress != address(0) &&
safetyFeeAmount > 0) {

payable(safetyPoolAddress).transfer(safetyFeeAmou
nt);
    }

    loan.repaid = true;

    // Increase borrower's trust score for successful
repayment
    uint256 currentScore =
trustScoreContract.getScore(loan.borrower);
    uint256 newScore = currentScore +
repaymentBonus;
    // Cap at 100
    if (newScore > 100) newScore = 100;
    trustScoreContract.updateScore(loan.borrower,
newScore);

    emit LoanRepaid(loanId, totalRepayment,
interest);
    emit TrustScoreIncreased(loan.borrower,
repaymentBonus);

    // Return excess payment if any
    if (msg.value > totalRepayment) {
        payable(msg.sender).transfer(msg.value -
totalRepayment);
    }
  }

  // ------------------------------
```

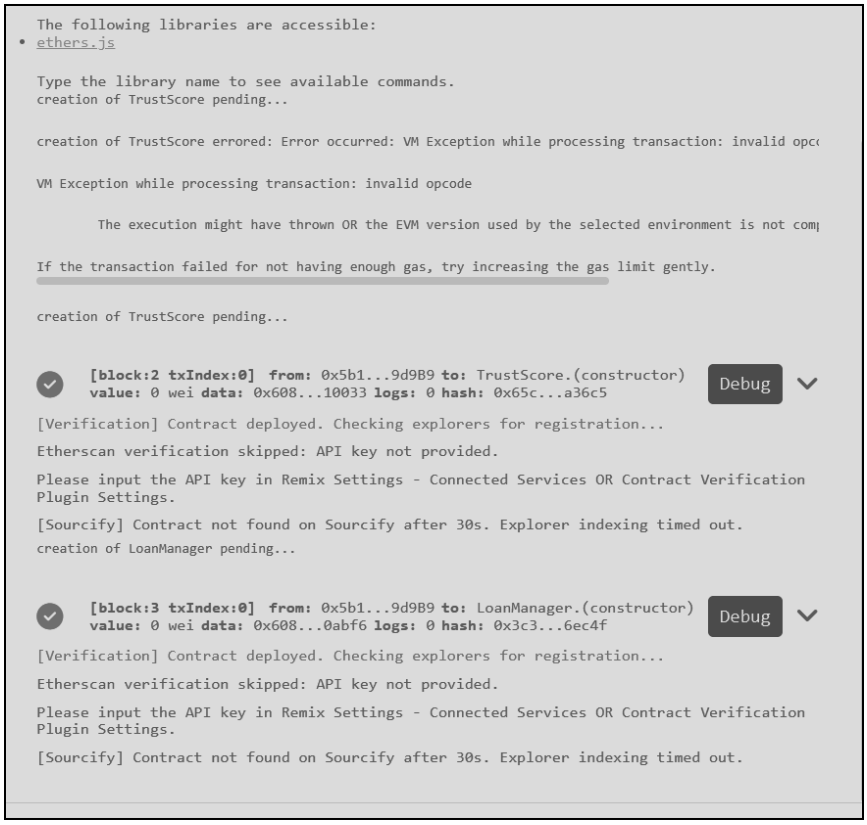
```
// GET ALL LOANS
// ------------------------------
function getLoans() public view returns (Loan[] memory) {
    return loans;
}
}
```



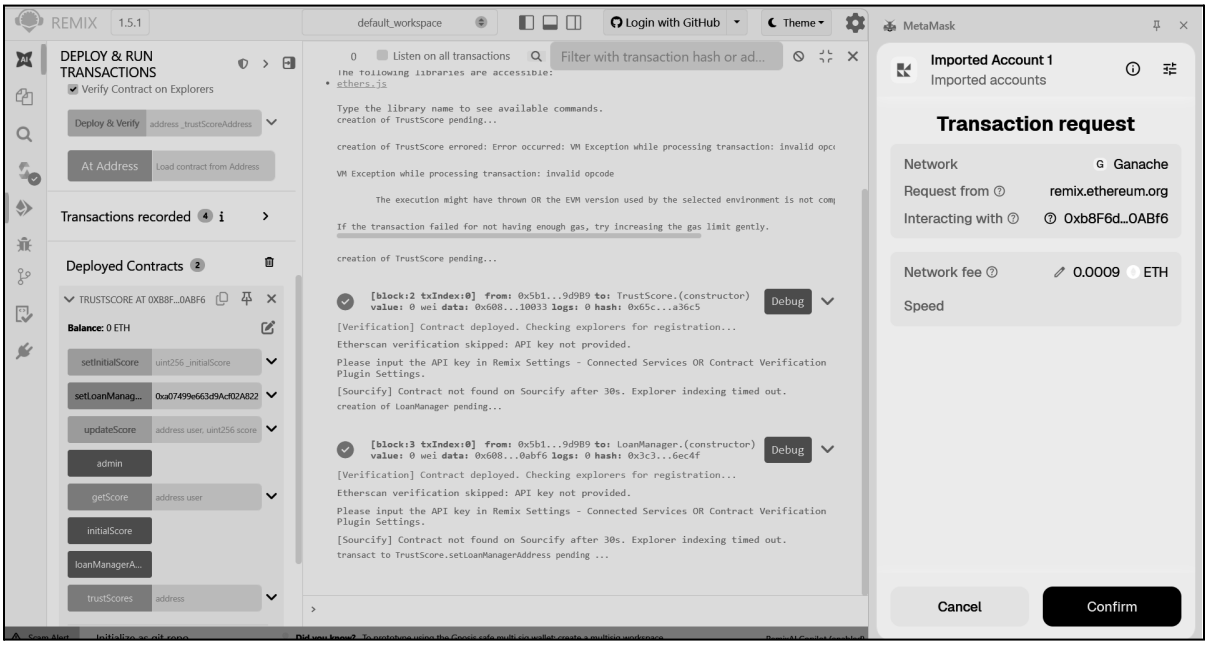


Contracts Deployed successfully-



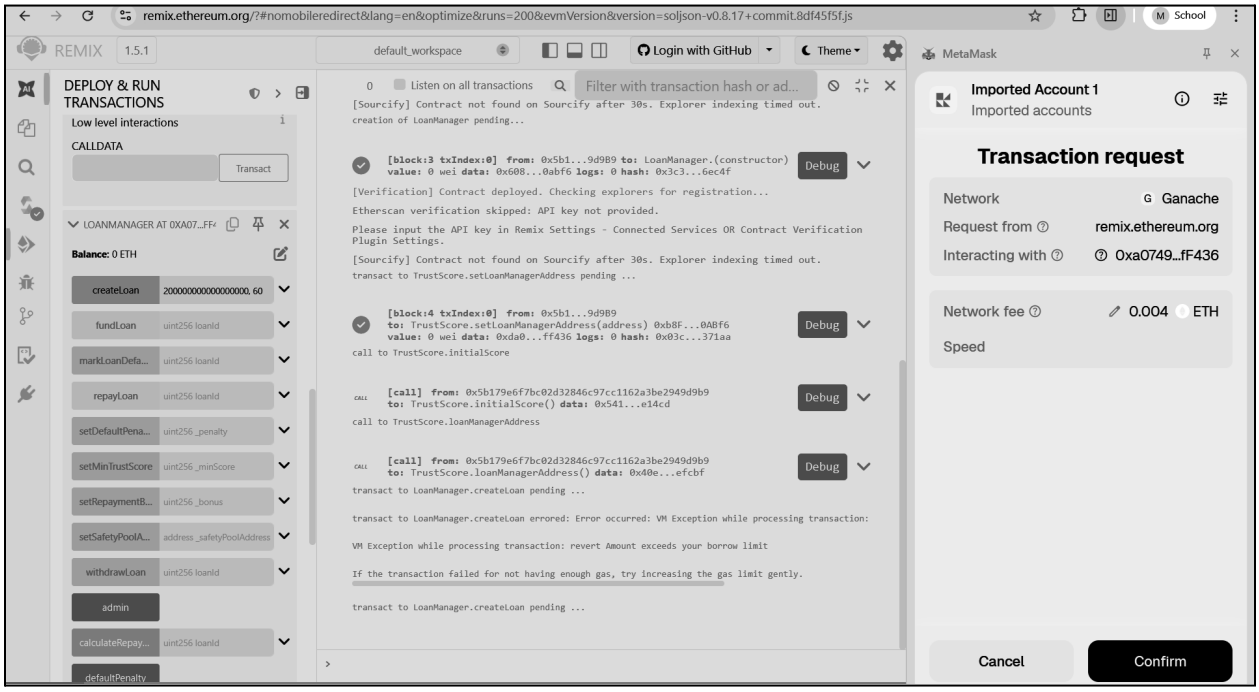


Linking both the contracts by function- setLoanManagerAddress(LoanManager_address)

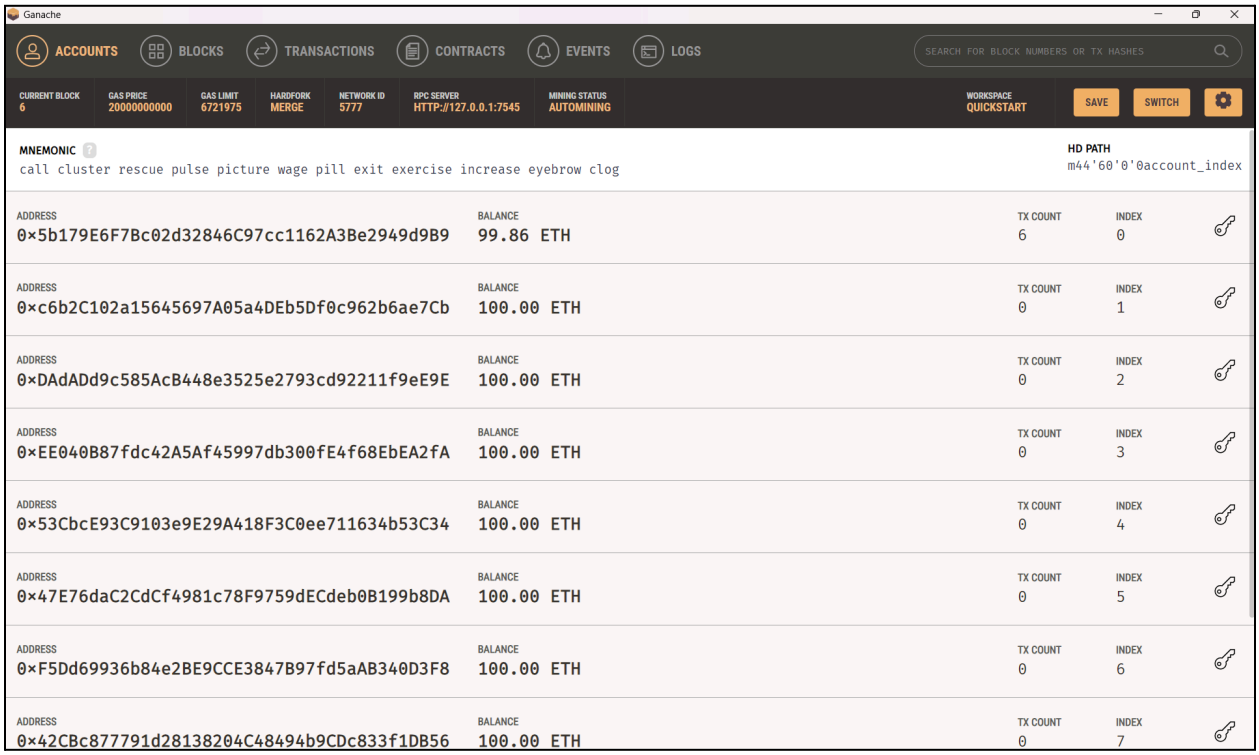


getScore(your_address) - is 50 which is default

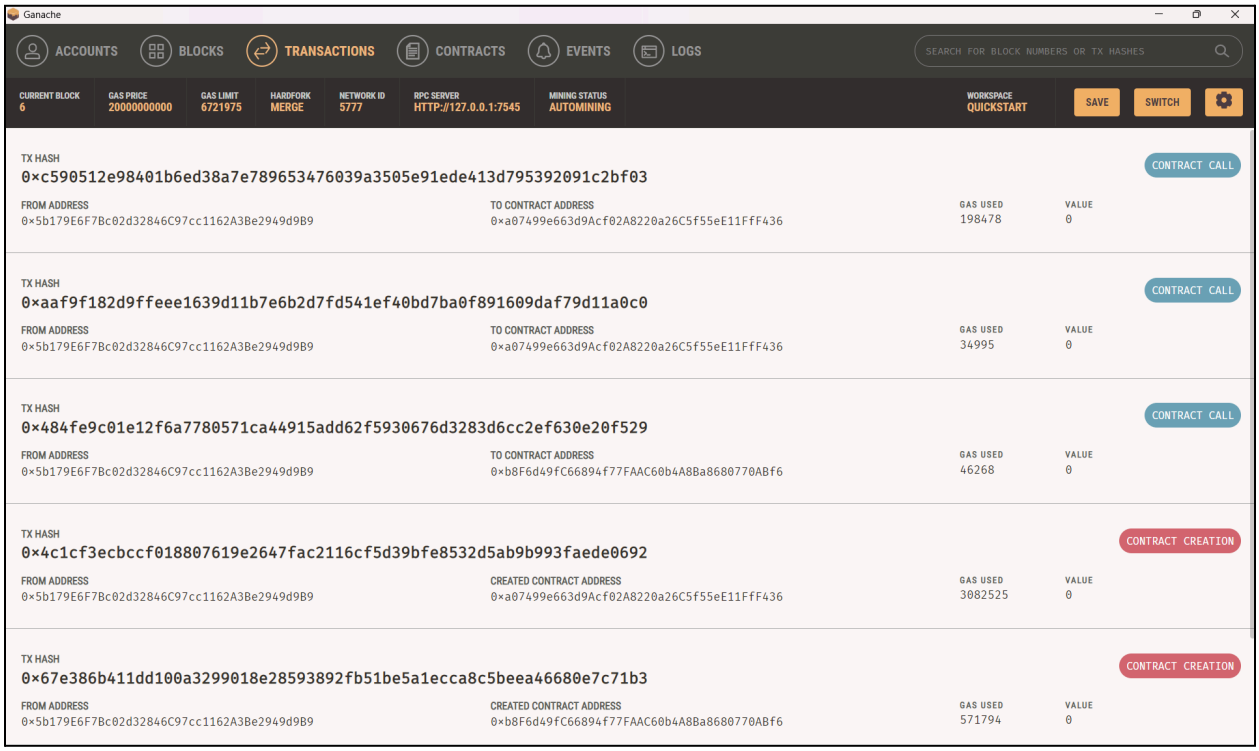
Function createLoan - Loan of 2 Eth created



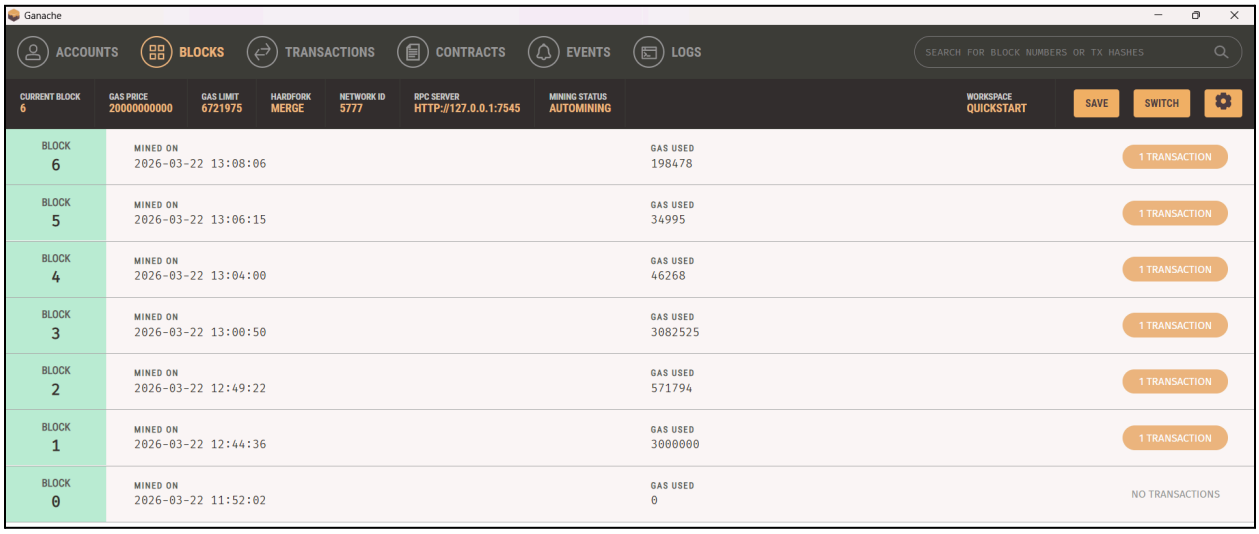
Balance became 99.86 ETH after all the transactions-



Transactions in Ganache-



No. of Blocks Mined-



Conclusion:

Ganache provides a simple and powerful environment to simulate a local Ethereum blockchain. By integrating it with **Metamask** and **Remix**, developers can deploy, test, and debug smart contracts efficiently before moving to a public network.